Fine-tuning GLiNER2.5-Decide on the National Bibliography of Scotland

Time estimates for Getty AAT / ULAN / TGN reconciliation · 26 October 2026 · Planning estimates: not yet measured on NBS data or EIDF hardware

Summary

- **Training a Yale-scale model** (about 25,000 labelled decisions, 2 epochs) takes about **1.5–2.5 hours on one EIDF H100 or H200**, 4–5 hours on an EIDF A100, and **12–20 hours on an RTX 5060 Ti 16 GB**.
- **Finding candidates is no longer a bottleneck.** The local Getty index answers in 5–30 ms per unique term (measured), so even the whole bibliography takes under 3 hours on a laptop. The live Getty service would take over a week.
- **Labelling is the real constraint.** 25,000 decisions is too many to review by hand; plan on language-model labels with a human-checked validation set, as the Yale team did.
- **Getty only partly covers a bibliography.** Places, genres and physical-description terms match well. Most authors are not in ULAN, so person terms will mostly be *none*.

Basis of the estimates

- **Reference run:** the Yale model GLiNER2.5-Decide-linkedart-recon trained on 24,318 decisions for 2 epochs (3,040 steps, effective batch 16, bf16, gradient checkpointing) in 283 minutes on one NVIDIA L40S. That is about **0.35 s per decision per epoch**, including periodic evaluation, or about 12 minutes per 1,000 decisions over 2 epochs.
- **Other GPUs** are scaled from that run by their bf16 compute and memory bandwidth relative to the L40S. These are not benchmarks; run `train.py --preflight-only` on the target hardware to confirm memory and speed.
- **NBS size:** 507,089 records (version 6, CC-BY 4.0, MARCXML). Terms per record are assumed at about 4 (place of publication, subjects, genre, names, physical description) and have not been measured.
- **Likely upper bound:** bibliographic records are shorter than museum descriptions, and Getty-only candidate lists are shorter (12.8 candidates on average in the sample, against about 22 in the Yale data), so each NBS example should cost less than a Yale example.

Hardware

GPU	Memory	Speed vs L40S	Batch setting	Notes
NVIDIA L40S (reference)	48 GB	1×	8 × 2 accumulation	Yale's run: 21 GiB peak on the longest batch
EIDF H200	141 GB	~2.2–3.3×	8 × 2	16 GPUs on the service
EIDF H100	80 GB	~2–3×	8 × 2	152 GPUs; usually the longest queue
EIDF A100 80 GB / 40 GB	80 / 40 GB	~1.0–1.3×	8 × 2	80 + 56 GPUs
EIDF A100 MIG slice 3g.20gb	20 GB	~0.35–0.55×	4 × 4	About half an A100 40 GB
EIDF A100 MIG slice 1g.5gb	5 GB	Not viable	—	A full fine-tune needs ~5.4 GB before any data
RTX 5060 Ti 16 GB	16 GB	~0.25–0.4×	2–4 × 4–8	448 GB/s; ~47 bf16 TFLOPS (third-party estimate)

Training time (2 epochs, one GPU)

Labelled decisions	Roughly	L40S (reference)	EIDF H100 / H200	EIDF A100	EIDF MIG 3g.20gb	RTX 5060 Ti 16 GB
2,000	Pilot, ~500 records	~25 min	~7–12 min	~18–23 min	~45–65 min	~1–1.5 h
25,000	Yale scale, ~6–8k records	~5 h	**~1.5–2.5 h**	~4–5 h	~9–14 h	**~12–20 h**
100,000	Large, ~25–30k records	~20 h	~6–10 h	~15–20 h	~1.5–2.3 days	~2–3.3 days
~2,000,000	Whole NBS at ~4 terms/record	~16 days	~5–8 days	~12–16 days	~4–6.5 weeks	~6–9 weeks

Training on the whole bibliography is not recommended: the Yale model reached 0.926 accuracy from 24,000 decisions, and gains beyond about 100,000 are likely to be small. If needed, EIDF's default quota of 12 GPUs allows multi-GPU training (supported by gliner2's trainer); 8 H100s would bring the whole-NBS row to about a day.

Data preparation and running the model

Stage	Estimate	Basis
Build the local Getty index (one-off)	~17 min	Measured: AAT 1 min, ULAN 2.5 min, TGN 12 min, search index 1.5 min
Find candidates: local index	5–30 ms per unique term 25k sample: under 5 min Whole NBS: ~25 min–2.5 h	Measured on 47 test terms; whole-NBS figure assumes ~300,000 unique terms
Find candidates: live Getty service	2–3 s per unique term 25k sample: ~6–8 h Whole NBS: over a week	Polite request rate; repeated terms are free from the cache
Labelling	The main constraint	LLM-assisted labels plus a human-checked validation set (1,000–2,000 decisions)
Running the trained model	~0.09 s per decision Whole NBS: ~2 days on one data-centre GPU, 1–2 weeks on a 5060 Ti	Yale: 1,224 decisions in 108 s

Practical notes

- **EIDF GPU Service:** jobs run as Kubernetes jobs queued by Kueue. The default project quota is 12 GPUs, 100 CPU cores and 1 TiB memory. Access is through the EIDF portal for University of Edinburgh researchers and DDI Programme partners.
- **RTX 5060 Ti:** use Linux or WSL2 with a PyTorch build for CUDA 12.8 or later (needed for Blackwell cards). Run the training script's preflight check first to find the largest batch that fits in 16 GB.
- **Getty snapshot:** the local index holds AAT from August 2026 and ULAN and TGN from January 2026. Against the live service it gave the same top candidate for all 47 test terms; the few other differences were records Getty edited after January 2026.
- **Starting model:** fastino/GLiNER2.5-Decide is Apache-2.0. Starting from the Yale model gives a head start, but the result inherits its CC BY-NC 4.0 licence.

Sources

Yale Cultural Heritage, GLiNER2.5-Decide-linkedart-recon model card and training code (Hugging Face) · National Library of Scotland Data Foundry, National Bibliography of Scotland v6 · EIDF GPU Service website and documentation · RTX 5060 Ti specifications (flopper.io, Guru3D) · GPU comparisons (Spheron, CUDO Compute) · Local measurements of the recon_prep Getty index, 23–26 October 2026.